Daily Iterative Learning Control With a Tuned PID Prior for Dissolved Oxygen Control in BSM1: Comparison With Multiagent Adaptive Critic Control

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***Abstract*—Wang, Li, Ren and Qiao, in "Multiagent Adaptive Critic Control With Expert Knowledge for Wastewater Treatment Plants" (IEEE Transactions on Industrial Informatics, 2026), reported reductions of 92 to 95 % in the dissolved oxygen tracking error of the Benchmark Simulation Model No. 1 (BSM1) relative to the PID controller used as expert knowledge. This paper proposes an alternative that requires no neural network: a PID with gains selected per term under a stability margin, a daily iterative learning control (ILC) feedforward with bumpless update and anti-windup, and a variant with a first-order measurement filter for noisy sensors. All tuning used the first seven days of the dry weather influent, and days 7 to 14 of the dry, rain and storm influent were used for evaluation. The proposed controller reduces the oxygen integral absolute error (IAE) by 90, 89 and 80 % relative to the PID prior of the multiagent adaptive critic controller (MAACC), with the same aeration energy. With the published parameters, MAACC remained within 1 % of its PID prior in all 80 runs; with replay-based twin-critic training it reached 85, 52 and 67 %. A soft actor-critic correction trained by imitation added no further improvement to the proposed controller. Additional tests with influent from BSM2 and with the BSM1 sensor models show that the improvement is retained for new influent after primary treatment (89 %) but not under measurement noise, where none of the evaluated controllers improved on the PID prior.**

***Index Terms*—Wastewater treatment, BSM1, dissolved oxygen control, iterative learning control, PID control, adaptive critic control, reinforcement learning.**

I. Introduction

AERATION accounts for the largest share of the energy used in activated sludge plants, and the dissolved oxygen (DO) loop that sets it is the primary control loop of such plants. The Benchmark Simulation Model No. 1 (BSM1) [11], [13] defines a common plant, influent data and performance measures for comparing control strategies.

Several recent studies apply learning controllers to BSM1 that start from a PID controller and learn a correction to it [1]–[3]. Wang *et al.* [1] proposed multiagent adaptive critic control (MAACC), in which one agent per loop combines an action network with an incremental PID used as expert knowledge, and reported oxygen tracking errors 95, 95 and 92 % lower than those of the PID in dry, rain and storm weather.

The size of such improvements depends on the tuning of the reference PID and on the evaluation conditions. The BSM1 influent has a strong daily pattern, and most of the error left by a PID follows this pattern. Iterative learning control (ILC) [15], [19] is designed for this type of repeated disturbance and does not require a model or training data.

The contributions of this paper are as follows.

1) A dissolved oxygen controller that combines a PID with gains selected per term and a daily ILC feedforward with bumpless update and anti-windup.
2) A noise-robust variant with a first-order measurement filter and reduced gains, selected using the BSM1 sensor models.
3) A comparison with MAACC [1] under a common protocol with separate tuning and evaluation periods, including a version of MAACC trained with replay and twin critics, and a soft actor-critic correction trained by imitation.
4) An evaluation on influent from BSM2, under actuator saturation and with the BSM1 sensor models, which shows the limits of all evaluated controllers under measurement noise.

II. Related Work

BSM1 [13] couples five ASM1 reactors [17] with a ten-layer settler and defines influent files, sensor and actuator models and evaluation criteria. BSM2 [14] extends it to a full plant with a 609-day influent that includes seasonal and weekly variation.

Adaptive critic designs belong to the adaptive dynamic programming family [12]. Wang *et al.* [1] use one adaptive critic agent per loop with a PID prior; Zhao *et al.* [2] combine online Q-learning with a PID; Du *et al.* [3] use deep reinforcement learning to adjust the oxygen set point. Deep actor-critic methods relevant here include twin-critic training [8], its multiagent form [6], [10], soft actor-critic [7] and imitation regularization [5]. Results of such methods are known to depend on random seeds, baselines and evaluation details [9].

ILC improves a repeated task from one trial to the next [15], [19]. The update used in this paper learns a feedforward from the correction made by the feedback controller, which is the principle of feedback-error learning [18].

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Code and result files: https://github.com/Amritha902/water_resource.

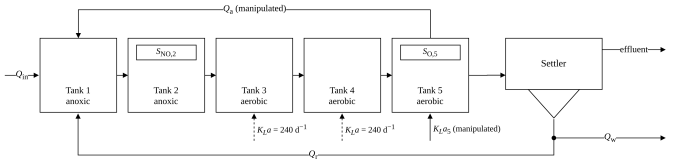


Fig. 1. BSM1 plant with the two manipulated variables of the control problem: the oxygen transfer coefficient of tank 5, $K_L a_{,5}$, and the internal recycle flow, Q_a.

III. Plant and Problem Formulation

The plant (Fig. 1) consists of two anoxic tanks of $1000\,\mathrm{m}^3$, three aerated tanks of $1333\,\mathrm{m}^3$ and a ten-layer secondary settler, simulated with the BSM1 implementation of `bsm2-python` 0.0.16 [4]. As in [1], two loops are closed with a sampling period of 45 s: the oxygen concentration in tank 5, $S_{O,5}$, is controlled to 2 mg/L through $K_L a_{,5} \in [0, 240]\,\mathrm{d}^{-1}$, and the nitrate concentration in tank 2, $S_{NO,2}$, to 1 mg/L through $Q_a \in [0, 92\,230]\,\mathrm{m}^3/\mathrm{d}$. The set points are printed as $[1, 2]$ in [1]; the published figures and PID errors correspond to the standard values $[2, 1]$, which are used here.

The PID prior of [1] is incremental:

$$\Delta u_k = K_p(e_k - e_{k-1}) + K_i e_k + K_d(e_k - 2e_{k-1} + e_{k-2}), \quad (1)$$

with $K_p = \mathrm{diag}\{100, 10^5\}$, $K_i = \mathrm{diag}\{20, 3{\times}10^4\}$ and $K_d = \mathrm{diag}\{10, 10^3\}$ for the oxygen and nitrate loops.

Performance is measured, following [1], by

$$\mathrm{IAE} = \frac{1}{N}\sum_{k=1}^{N} |e_k|, \quad (2)$$

computed from the true concentrations. The aeration movement $\sum_k |K_L a_{,5}(k) - K_L a_{,5}(k-1)|$ per day, the aeration energy and the effluent ammonium are also reported. A run is terminated if $S_{O,5}$ leaves the range 0.3 to 4 mg/L.

The influent consists of the BSM1 dry, rain and storm files (14 days, 15-min samples). In the rain and storm files, the inert soluble COD is diluted as $S_I = 30\,Q_{\mathrm{dry}}/Q$, as in the official rain data. Every run starts from the state reached after 14 days of dry weather under the PID prior.

IV. Baseline: Multiagent Adaptive Critic Control

In MAACC (Fig. 2a), each loop has an agent whose action network outputs an increment Δa_i, which is integrated and added to the PID output, $u_i = a_i + b_i$. A critic network that receives both loops estimates the cost-to-go and is updated by a one-step temporal-difference error, and the action network follows the gradient of the critic. The equations and the weights and learning rates of Table I in [1] were implemented without modification and evaluated in 20 runs for each of four cases (dry, rain, storm and dry with actuator noise).

Table I shows that the PID results agree with [1] within 1 % in dry and storm weather and within 10 % in rain. For the nitrate loop, the PID error obtained here is approximately half of the published value (0.0064 against 0.0145 in dry weather). MAACC with the published parameters remained within 1 % of the PID in all 80 runs: with the published initial weights, the gradient reaching the action networks is of the order of 10^{-4}, and with learning rates of 0.01 and 0.005 the networks change little over 14 days. Twelve alternative initializations either remained at the PID or diverged.

TABLE I
Oxygen IAE over 14 days: values published in [1] and values obtained in this study.

	Dry	Rain	Storm
PID, published	0.0114	0.0098	0.0117
PID, this study	0.0114	0.0108	0.0118
MAACC, published	0.000527	0.000509	0.000947
MAACC, this study	0.0114	0.0108	0.0118

V. Proposed Controller

A. PID Tuning

The P and I multipliers of each loop were selected from $\{1, 2, 3, 4, 6, 8\}$ (36 combinations per loop, D multiplier equal to P). A combination was admissible only if the loop remained within the operating range on the tuning data with all its multipliers increased by a further factor of 1.5. The admissible combination with the lowest IAE on dry days 0 to 7 is P×2, I×8, D×2 for oxygen and ×3 for nitrate. The oxygen selection lies at the edge of the grid. A uniform gain factor of 3 (PID×3) is also reported as an intermediate step.

B. Daily Iterative Learning Control

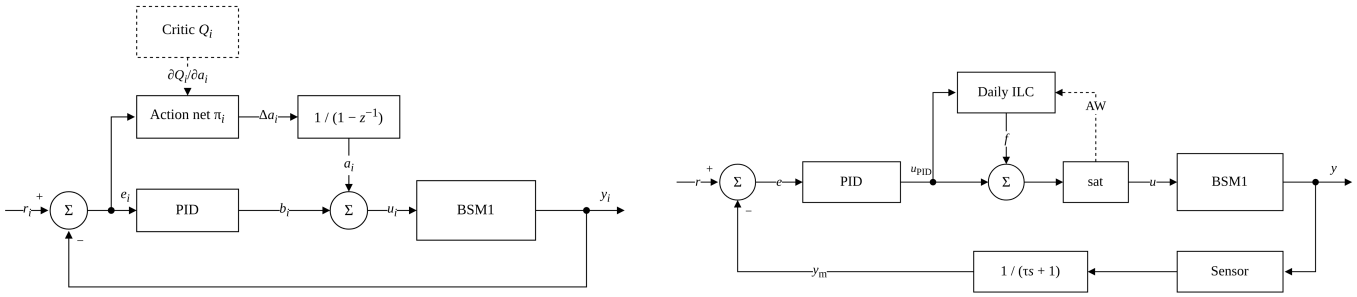
The BSM1 influent follows a daily pattern, and the PID responds only after the morning load increase has displaced the oxygen concentration. Each day is therefore treated as one trial. With $j \in \{1, \ldots, 1920\}$ the 45-s slot of the day, the applied aeration is $u_k = \mathrm{sat}(u_k^{\mathrm{PID}} + f_d(j_k))$, and at the end of day d the feedforward profile is updated as

$$f_{d+1} = \mathcal{S}_w\left[f_d + \beta\,\mathcal{L}_\ell\left(u_d^{\mathrm{PID}} - \bar{u}_d^{\mathrm{PID}}\right)\right], \quad (3)$$

where $\mathcal{L}_\ell$ advances the recorded profile by ℓ slots, $\mathcal{S}_w$ is a circular moving average over $\pm w$ slots and $\bar{u}_d^{\mathrm{PID}}$ is the daily mean. The ILC thus transfers the recurring part of the PID correction to the feedforward [18]. Algorithm 1 gives the complete procedure, including two additions:

- *Bumpless update.* When the profile changes at midnight, the PID state is reduced by the same amount, so that the applied aeration is continuous. This reduced the largest oxygen deviation near midnight from 0.053 to 0.011 mg/L.
- *Anti-windup.* The PID limits are shifted by the feedforward, slots in which the actuator is saturated are excluded from the update, and a day saturated for more than half of its slots keeps the previous profile.

The ILC acts on the oxygen loop only. The parameters were selected on dry days 0 to 7 as the lowest IAE among settings whose largest oxygen deviation does not exceed that of the PID alone: $\beta = 1$, $\ell = 4$ (3 min) and $w = 30$ for the tuned PID, and $\beta = 0.75$, $\ell = 8$, $w = 30$ for PID×3.



(a) (b)

Fig. 2. (a) One agent of MAACC [1] ($i = 1$: oxygen, $i = 2$: nitrate). The critic receives the states and actions of both loops. (b) Proposed oxygen loop: tuned PID, daily ILC with anti-windup (AW), actuator saturation and the first-order measurement filter of the noise-robust variant.

Algorithm 1 Daily ILC with bumpless update and anti-windup

Require: β, ℓ, w; $f(j) \leftarrow 0$ for all slots j

for each control step k with slot j **do**
  $u_k^{\mathrm{PID}} \leftarrow$ (1) limited to $[U_{\min} - f(j), U_{\max} - f(j)]$
  $u_k \leftarrow \mathrm{sat}(u_k^{\mathrm{PID}} + f(j))$
  $g(j) \leftarrow u_k^{\mathrm{PID}}$; $o(j) \leftarrow 1$ if u_k is not saturated, else 0
  if $j = 1920$ and $\sum_j o(j) \geq 960$ **then**
    $\delta(j) \leftarrow o(j)\,\big(g(j) - \bar{g}\big)$, $\bar{g}$ the mean over $o(j) = 1$
    $f^+ \leftarrow \mathcal{S}_w[f + \beta\,\mathcal{L}_\ell\,\delta]$
    PID state $\leftarrow$ PID state $-(f^+(1) - f(1))$
    $f \leftarrow f^+$
  end if
end for

TABLE II
TRAINING SETTINGS OF THE LEARNED CORRECTIONS.

	SAC-IL	rMAACC
Discount factor	0.97	0.97
Learning rate	3×10^{-4}	3×10^{-4}
Batch; buffer size	256; 4×10^5	256; 4×10^5
Target update rate	0.005	0.005
Bound per step	$6\,\mathrm{d}^{-1}$	$6\,\mathrm{d}^{-1}$; $3000\,\mathrm{m}^3/\mathrm{d}$
Initialization	behaviour cloning	zero last layer, one day of exploration
Exploration	entropy, $\alpha_0 = 0.005$	Gaussian, std. 0.1
Imitation weight	2 decaying to 0.5	none
Loops corrected	oxygen	oxygen and nitrate

C. Noise-Robust Variant

With the BSM1 sensor models, the measured oxygen carries white noise with a standard deviation of 0.25 mg/L. A first-order filter $1/(\tau s + 1)$ was added to the oxygen measurement and the gains of both loops were scaled, with both parameters selected on dry days 0 to 7 using the sensor models and noise seeds that were not used for evaluation. The selected values are $\tau = 0.75$ min and a gain scale of 0.5, combined with the ILC with anti-windup (Fig. 2b).

D. Learned Corrections

Two learned corrections were evaluated on top of PID priors under the same data budget (dry days 0 to 7, five episodes of seven days, five training runs each); Table II lists their settings.

- *MAACC with off-policy training (rMAACC).* The structure of MAACC is kept, and the training is replaced by that of multiagent TD3 [6], [8]: replay buffer, target networks, twin critics, target-policy smoothing and delayed actor updates. The last layer of the action network is initialized to zero, so that training starts from the PID.
- *SAC-IL.* A soft actor-critic [7] adds a bounded increment to the aeration. Its actor (18 564 weights) receives the last three errors of both loops, both inputs and the influent flow; training starts with behaviour cloning of the PID with ILC, and an imitation term with a lower bound is kept in the actor loss [5]. The reward is $r = -(100|e_O| + 50|e_N|) - 0.05\|a\|^2$. The learned part is disabled when an input leaves its training range by more than 10 %.

VI. EXPERIMENTAL SETUP

All tuning and training used dry weather days 0 to 7. Days 7 to 14 of the dry, rain and storm files form the test period; the rain and storm events occur only in this period. The dry file was also run for 14 days with the actuator noise of [1] (standard deviation 3 % of each upper bound).

The following additional conditions were evaluated, each scored on days 7 to 14 of a 14-day run:

- H1, H5: six 14-day windows of the BSM2 influent [14] each (start days 245 to 525 and 273 to 553), flow scaled by 0.878 to the BSM1 dry mean. Without primary treatment this influent saturates the aeration for about 30 % of the time.
- H4: the H5 windows with the particulate components halved, as a simplified primary clarifier.
- H2, H6: the dry, rain and storm files with the BSM1 sensor models [13] (oxygen: $T_{90} = 1$ min; nitrate: $T_{90} = 10$ min; noise standard deviation 0.25 mg/L), three noise seeds each.
- H3: as H2 with sensor delay only.
- H7: H4 with the sensor models.

The anti-windup and the noise-robust variant were developed after conditions H1 to H3 had been evaluated, and are therefore

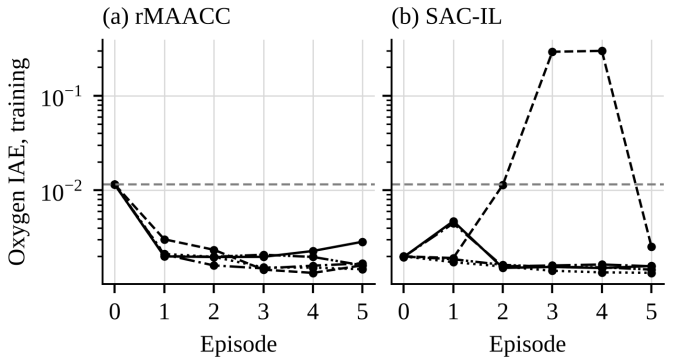


Fig. 3. Oxygen IAE on the training period (dry days 0 to 7) after each training episode, five independent training runs per method, from a complete retraining. Episode 0 is the initial policy: the PID prior for rMAACC and behaviour cloning of PID×3 with ILC for SAC-IL. Dashed line: PID prior.

assessed on H4 to H7 only. Percentages compare a controller with the PID prior over the runs completed by both.

VII. Results

A. Training of the Learned Corrections

Fig. 3 shows the training-period oxygen IAE of both learned corrections over five training runs each. rMAACC improves on its PID prior within the first episode, from 0.0115 to between 0.0020 and 0.0030; the retained policies reach 0.0013 to 0.0023. SAC-IL remains close to its imitation starting point (about 0.0020), and one of its five runs diverged in episodes 3 and 4 before recovering. In each run, the policy with the lowest combined oxygen and nitrate error on the training period was retained. Retraining from scratch with the same seeds reproduced the training curves of all ten runs to five decimal places.

B. Test Period

Table III summarizes the test period. The gain tuning accounts for most of the improvement (77 to 85 % for the tuned PID), and the ILC adds 3 to 5 points on the tuned PID and 12 to 19 points on PID×3. SAC-IL changes the result by at most 2 points. Anti-windup leaves these results unchanged, since the aeration is rarely saturated in this period. rMAACC reaches the level of the tuned PID in dry weather only and moves the aeration 3.7 times as much as the PID prior. The aeration energy (3690 kWh/d) and effluent ammonium (2.47 g N/m^3) are equal for all controllers to the reported precision. For the nitrate loop, the controllers based on PID×3 or the tuned PID reduce the IAE by 73, 46 and 34 %, and the noise-robust variant by 36, 28 and 15 %; rMAACC reduces it by 62 and 5 % in dry and rain weather and increases it by 6 % in storm weather.

Fig. 4 shows the mechanism. The PID prior lets the oxygen concentration fall at each morning load increase, whereas the proposed controller holds it at the set point. The required change in aeration is small (at most 6 d^{-1}) and occurs at the load increase.

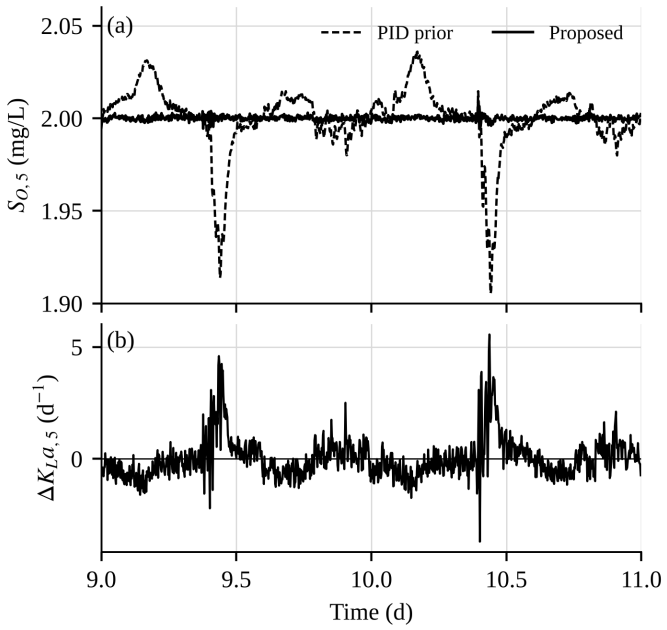


Fig. 4. Two days of the dry test period. (a) Oxygen concentration under the PID prior (dashed) and the proposed controller (solid). (b) Difference in $K_L a_{,5}$ between the two controllers.

C. Additional Conditions

Table IV gives the results for the additional conditions.

- *New influent.* With primary treatment (H4), the proposed controller retains its improvement (89 %). Without it (H1, H5), the aeration saturates, and the ILC without anti-windup increases the error by 48 to 76 %, because the feedforward learned during saturation is applied when saturation ends. Anti-windup reduces the error under saturation by 39 % relative to the ILC without it (lower in five of six windows), to the level of the tuned PID.
- *Sensor delay.* With delay only (H3), rMAACC gives the largest improvement (86 %), followed by the tuned PID (74 %).
- *Measurement noise.* With the BSM1 sensor noise (H2, H6, H7), the higher-gain controllers either left the operating range or performed worse than the PID prior. The noise-robust variant completed all runs but was 3 and 24 % worse than the PID prior on H6 and H7, and rMAACC was within 6 %. Under noise the aeration reaches its limits frequently; with anti-windup the ILC then stops learning, and the controller behaves as the tuned PID alone, which does not tolerate the noise.

VIII. Discussion

Learning in MAACC. An adaptive critic updated once per step from that step alone, with a gradient of the order of 10^{-4} and learning rates of 0.01 and 0.005, does not move appreciably from its initial policy within 26 879 steps. Initialized from the PID, it therefore remains at the PID. With data reuse and stable targets, the same architecture improves the tracking in dry weather, which indicates that the difference between the published and the present results is more likely due to training details not reported in [1] than to the architecture.

TABLE III

TEST PERIOD (DAYS 7 TO 14): CHANGE IN OXYGEN IAE RELATIVE TO THE PID PRIOR. FIRST ROW: ABSOLUTE IAE OF THE PID PRIOR. ACTUATOR NOISE: DRY WEATHER, 14 DAYS. MOVEMENT: $K_La_{,5}$ MOVEMENT IN d^{-1} PER DAY, DRY WEATHER. LEARNED CORRECTIONS: MEAN OF FIVE TRAINING RUNS.

Controller	Dry	Rain	Storm	Actuator noise	Movement
PID prior [1] (IAE)	0.0113	0.0100	0.0120	0.0249	501
MAACC [1]	0 %	0 %	0 %	0 %	501
rMAACC	−85 %	−52 %	−67 %	−13 %	1835
PID×3	−66 %	−67 %	−60 %	−26 %	966
PID×3 + ILC	−85 %	−82 %	−72 %	−28 %	961
PID×3 + ILC + SAC-IL	−87 %	−82 %	−74 %	−26 %	982
Tuned PID	−85 %	−85 %	−77 %	−22 %	852
Tuned PID + ILC (proposed)	−90 %	−89 %	−80 %	−23 %	852
Tuned PID + ILC + SAC-IL	−90 %	−88 %	−80 %	−22 %	874
Noise-robust variant (proposed)	−83 %	−80 %	−73 %	+10 %	522

TABLE IV

ADDITIONAL CONDITIONS, DAYS 7 TO 14: CHANGE IN OXYGEN IAE RELATIVE TO THE PID PRIOR OVER THE RUNS COMPLETED BY BOTH. FIRST ROW: ABSOLUTE IAE OF THE PID PRIOR. NUMBER OF TERMINATED RUNS IN PARENTHESES WHERE NONZERO; –: NO COMPLETED RUN OR NOT EVALUATED.

Controller	H1	H2	H3	H4	H5	H6	H7
PID prior (IAE)	0.106	0.255 (2/9)	0.0319	0.0105	0.0815	0.261	0.178
MAACC	–	–	–	+43 % (1/6)	−59 % (4/6)	–	–
rMAACC	−6 %	+1 % (2/45)	−86 %	−73 %	−8 %	+1 % (5/45)	+6 % (1/30)
PID×3 + ILC + SAC-IL	+52 %	–	−56 %	−77 %	+33 %	–	+144 % (25/30)
Tuned PID	−7 %	+25 % (8/9)	−74 %	−85 %	−10 %	–	+84 % (5/6)
Tuned PID + ILC	+76 %	+46 %	−73 %	−89 %	+48 %	+46 %	+50 %
Tuned PID + ILC with anti-windup	–	–	–	−89 %	−10 %	–	+84 % (5/6)
Noise-robust variant	–	–	–	−83 %	−10 %	+3 %	+24 %

Role of the ILC. The error that a PID leaves on BSM1 is dominated by its lag behind the daily load pattern. A daily ILC addresses this component directly, with a computational cost of about 1 μs per step and without training data. After this component has been removed, a learned correction has little remaining error to act on, which is consistent with the negligible contribution of SAC-IL on the tuned PID with ILC.

Measurement noise. Gain increases that reduce the error with ideal measurements amplify measurement noise [16]. With a noise standard deviation of 0.25 mg/L on a set point of 2 mg/L, this effect outweighs the improvement in tracking. Comparisons of controllers on BSM1 should therefore include the sensor models of the benchmark, an influent not used for tuning, and the actuator movement in addition to the tracking error.

Computation. The PID requires 5 μs per step, the MAACC update 37 μs and the SAC actor 35 μs, all far below the 45-s sampling period. One SAC-IL training run takes about 15 min on one processor core.

Limitations. The comparisons use the PID prior as implemented here, which matches the published oxygen error but not the published nitrate error. The termination criterion, the S_I correction, the train and test split, the simplified primary clarifier and the sensor noise generator (one draw every 45 s) are choices of this study. PID-based controllers are deterministic and were run once per condition; the learned corrections were trained five times. With six windows or nine noise runs per condition, means and counts are reported without significance tests. The test period was examined before the bumpless update was introduced and before the tuned PID was added.

IX. CONCLUSION

A PID with gains selected per term and a daily ILC with bumpless update and anti-windup reduces the oxygen IAE of BSM1 by 80 to 90 % relative to the PID prior of MAACC on a test period excluded from tuning, without a neural network and with the same aeration energy. Under the same protocol, MAACC with the published parameters remains equal to its PID prior, and learned corrections do not improve on the proposed controller. With the BSM1 sensor noise, none of the evaluated controllers improves on the PID prior; a controller that combines the tracking of the proposed method with robustness to measurement noise remains to be developed.

DATA AVAILABILITY

The code, result files and scripts that produce every table and figure are available at https://github.com/Amritha902/water_resource. A complete rerun of the training of both learned corrections and of the 138 test-period evaluations of the PID prior, MAACC, rMAACC, SAC-IL and the proposed controllers reproduced all stored results exactly.

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